

ÁLVARO GALVÁN

Geospatial engineer · Wildfire and climate risk

San Francisco, CA
alvaro.galvansandoval2001@gmail.com
[my-portafolio-chi.vercel.app](#)

EXPERIENCE

PANO AI

Jun 2025 – Aug 2025, May 2026 – Present

Geospatial Analyst · San Francisco, CA

- Migrated core viewshed and coverage computations from local ArcGIS to GCP, cutting processing time 10x and enabling scoring across 1,000+ infrastructure assets.
- Owned Pano's wildfire risk-scoring algorithm, developing the scoring metrics and network analysis behind customer-facing proposals for where to site cameras on real terrain.
- Extended the scoring engine from lines and polygons to point geometries, implementing inverse-distance and point-count methods in Python calibrated against hands-on work with the cameras themselves.
- Established the geospatial team's engineering practice in a codebase that had none: required PR review, CI checks, and the team's first test suite.

FUEGO.EARTH

Sep 2025 – Apr 2026

Geospatial Full-Stack Software Engineer · San Francisco, CA

- Co-built a cloud-native wildfire simulation platform as one of two engineers, after finding no free tool that could run and compare existing fire models.
- Integrated six external weather and geospatial APIs, LANDFIRE, NOAA, OpenMeteo, Copernicus, Sentinel, and NASA VEDAS, to feed C++ fire-behavior models, FlamMap and Cell2Fire.
- Deployed and maintained the platform end-to-end, React/TypeScript frontend on Vercel and backend on Hetzner, processing 1,000+ fire-spread simulations daily.
- Built the operator-facing visualization layer in React and D3, rendering fire progression as time-stepped isochrones so crews could read spread and risk.
- Validated simulated fire perimeters against satellite-observed burn scars before public release.

GRIDWARE

May 2023 – Nov 2024

Data Analyst · San Francisco, CA

- Investigated 5,000+ anomaly alerts and confirmed 300+ real events, maintaining 95%+ reporting accuracy.
- Designed real-time monitoring dashboards in Grafana, turning high-volume grid sensor data into clear visual signals for a 24/7 distributed operations team.

PROJECTS & VOLUNTEER WORK

Wildfire Risk Index — Hyticos (tiger conservation nonprofit, Telangana, India)

Built a continuously updating fire-index map in Python for teams with no capacity to assess fire risk themselves, fusing Sentinel-2 imagery with field-reported conditions and weighting factors by analytic hierarchy process.

Amazon Deforestation Classifier

Built a supervised classification pipeline in Python (Rasterio, GeoPandas, scikit-learn) to detect forest loss in the Brazilian Amazon from Sentinel-2 multispectral imagery, benchmarking Random Forest, XGBoost, and SVM to 90%+ accuracy.

EDUCATION

Minerva University

B.S. Computational Sciences, May 2026
San Francisco, CA

Coursework in software engineering, algorithms and data structures, machine learning, simulation and modeling of complex systems, optimization methods, networks, and Bayesian statistics.

SKILLS

LANGUAGES

Python, TypeScript, C++ (Basic)

CLOUD & FULL-STACK

GCP, SQL, React, dbt

GEOSPATIAL & ML

GeoPandas, PostGIS, Rasterio, Shapely, scikit-learn

DEVELOPER TOOLS

Git, GitHub, CI/CD, Cursor

INTERESTS

Wildfire and climate resilience, cartography, running

LINKS

LinkedIn [linkedin.com/in/alvarogalvan](#)

GitHub [github.com/AlvaroGalvan1](#)